

Anti-Hebbian plasticity protects against memory interference

Hebbian learning strengthens synaptic weights between correlated neurons and is believed to support assembly formation and pattern completion. Anti-Hebbian learning, on the other hand, strengthens connections between negatively correlated neurons, while its contribution to memory processing is unclear. Here we show in theory and experiment how anti-Hebbian plasticity at excitatory synapses onto inhibitory neurons (E-to-I synapses) protects against interference between competing memories. In the hippocampus, anti-Hebbian plasticity at E-to-I synapses is mediated by calcium-permeable AMPA receptors (CP-AMPA) that allow calcium influx due to presynaptic activity despite post-synaptic hyperpolarization. Using a viral approach targeting hippocampal parvalbumin-positive interneurons, we replaced CP-AMPA with calcium-impermeable receptors and observed a sharp decrease in anti-Hebbian plasticity. We developed an excitatory-inhibitory network model, where anti-Hebbian plasticity at E-to-I synapses allowed the storage of the negative part of a pattern covariance matrix, while the positive part was stored via Hebbian plasticity. When presented with a recall cue, the network converged to the cued pattern and maintained high accuracy despite strong cue noise or large pattern overlaps. Interestingly, when the strength of anti-Hebbian synapses was reduced, recall performance remained high for low cue noise and small pattern overlaps, but deteriorated quickly when noise or overlaps increased. These predictions were confirmed in CP-AMPA-deficient mice that performed indistinguishably from controls in a spatial memory task when delays between an encoding and a recall phase were short (corresponding to low cue noise in the model) but showed decreased performance for longer delays (high cue noise). Similarly, CP-AMPA-deficient mice learned the location of an escape platform in the Morris water maze but performed poorly when the platform's location was changed; indicating interference between highly similar memory patterns. Together, these results suggest that CP-AMPA-mediated anti-Hebbian plasticity at E-to-I synapses protects against interference between competing memory patterns by facilitating the storage of negative correlations.

Hebbian plasticity of E-to-E synapses has been suggested to be key for storing *positive* correlations between elements of a memory.¹ Hebbian plasticity in I-to-E and E-to-I synapses has been suggested to restore E-I balance during memory formation.² Here, we study *anti*-Hebbian plasticity in E-to-I synapses and show that it can store *negative* correlations between elements of a memory, and thereby support memory recall from ambiguous cues. Previous memory networks considered inhibitory interactions to maintain a global balance between excitation and inhibition. In these models, inhibitory neurons provide stabilizing feedback but do not directly contribute to memory storage.^{3–5} In contrast, experimental evidence suggests a role for inhibition beyond balancing excitation.^{6–8}

We consider E-I rate networks with dynamics

$$\tau \dot{\mathbf{r}} \propto -\mathbf{r} + [\mathbf{W}\mathbf{r}]_+^{\text{on}}, \quad [\mathbf{x}]_+ = \max(\mathbf{x}, 0), \quad (1)$$

$$\mathbf{r} = \begin{pmatrix} \mathbf{r}_E \\ \mathbf{r}_I \end{pmatrix}, \quad \mathbf{W} = \begin{pmatrix} \mathbf{W}_{EE} & -\mathbf{W}_{EI} \\ \mathbf{W}_{IE} & -\mathbf{W}_{II} \end{pmatrix}, \quad (2)$$

where bold symbols are matrices and vectors, and $[\cdot]_+^{\text{on}}$ denotes an element-wise power. For synapses w_{ij}^{+AB} connecting neurons i, j of type $A, B \in \{E, I\}$, we assume a synapse-type-specific competitive Hebbian learning rule⁹ with a BCM-like plasticity threshold¹⁰

$$\dot{w}_{ij}^{+AB} \propto (r_i^A - \bar{r}_i^A) r_j^B, \quad \sum_j w_{ij}^{+AB} = \omega_{AB}^+ \quad (3)$$

where r are firing rates with means $\bar{r}$ and synapses compete for type-specific resource pools such that total synaptic weights ω_{AB}^+ remain constant.

We assume that during memory storage, the network is dominated by memory patterns, $\mathbf{r}_A = \mathbf{p}$. Then

recurrent weights become proportional to the row-normalized positive part of the pattern covariance matrix $\bar{\mathbf{C}}^+$, while synapses that connect negatively correlated neurons decay to zero, maintaining Dale's law. Negative correlations are instead captured by anti-Hebbian learning, corresponding to CP-AMPA-mediated plasticity at E-to-I synapses^{11,12} (Fig. 1A):

$$\dot{w}_{ij}^{-IE} \propto -(r_i^I - \bar{r}_i^I) r_j^E, \quad \sum_j w_{ij}^{-IE} = \omega_{IE}^- \quad (4)$$

After memory storage, recurrent weights become

$$\mathbf{W} = \mathbf{W}^+ + \beta \mathbf{W}^- = \mathbf{W}^+ \otimes \bar{\mathbf{C}}^+ + \beta \mathbf{W}^- \otimes \bar{\mathbf{C}}^-, \quad (5)$$

where ' $\otimes$ ' is the Kronecker product, and $\mathbf{W}^+, \mathbf{W}^-$ hold type-specific weight norms $\omega_{AB}^+, \omega_{IE}^-$. To simulate a change in the strength of anti-Hebbian synapses, we introduced the scalar factor β .

Under some simplifying assumptions, we can show analytically that stored patterns become autonomous fixed points of the E-I network. Numerically, the network converges to the closest pattern when initialized with a noisy memory cue (Fig. 1B; C & D, top). When the strength of anti-Hebbian weights is decreased, the model predicts decreased recall performance when cue noise is large (Fig. 1C & D, bottom; E & F), but maintained performance when cue noise is small (Fig. 1E). We tested these predictions in PV-Cre mice deprived of CP-AMPA via bilateral (dorsal and ventral) injection of a GluA2 virus, which decreases anti-Hebbian plasticity (Fig. 1G). In a delay-dependent memory task (Fig. 1H), animal performance was only decreased when the delay between memory encoding and recall was long, while performance for shorter delays was similar

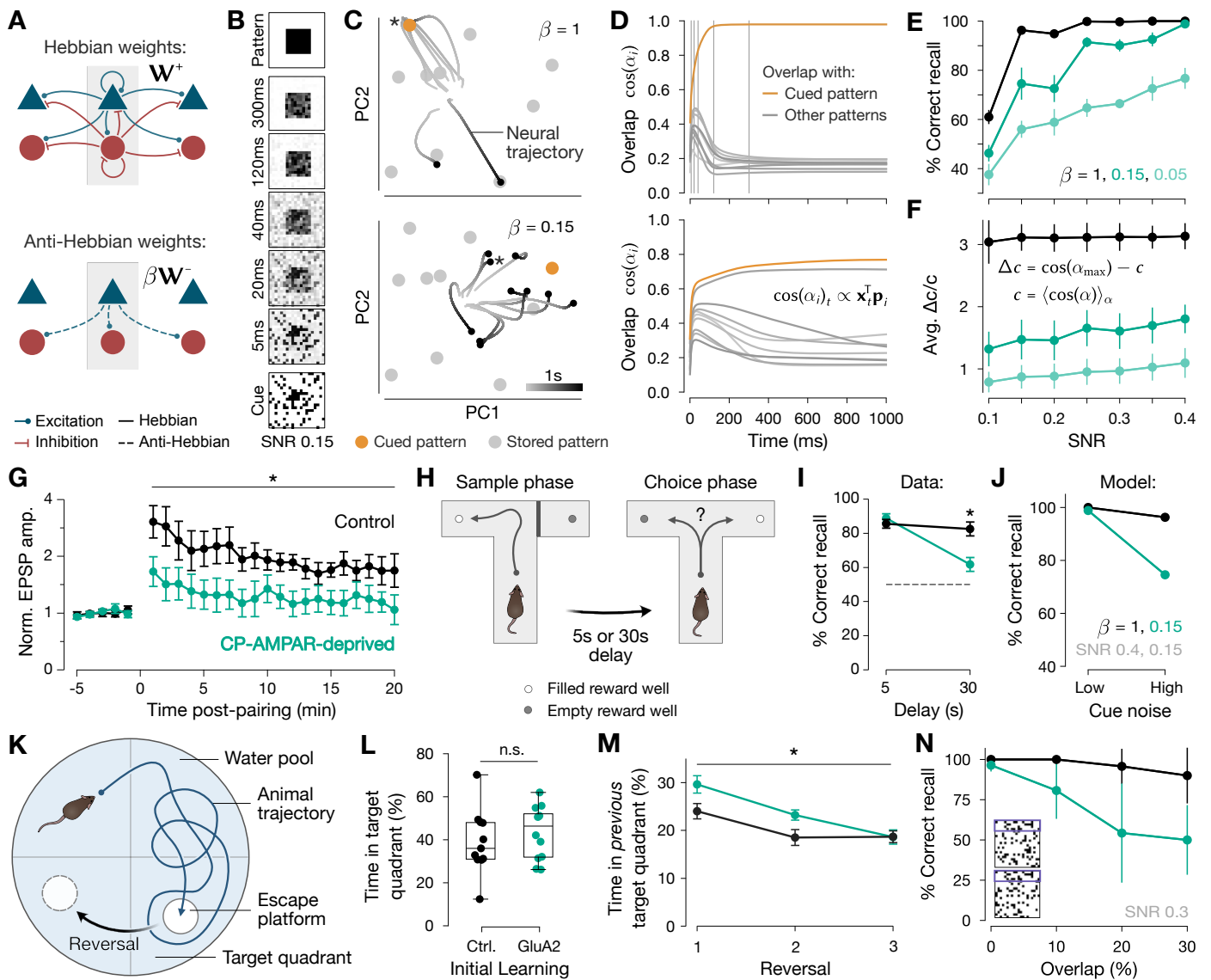


Figure 1. (A) Construction of E-I network. Only outgoing synapses from center E-I pair shown. (B) Neural activity $\mathbf{x}_E$ converging from noisy cue with low signal-to-noise ratio (SNR, bottom) to stored binary pattern (top) (cf. D, top). Pixels sorted to form a square for better visibility. (C) Neural trajectories $\mathbf{x}_E(t)$ in the principal pattern subspace. For large β , the network converged to stored patterns (top). For smaller β , recall accuracy decreased (bottom). Stars '*' indicate example trajectories in D. (D) Overlap (cosine similarity) of excitatory neural activities with cued (orange) and non-cued patterns (gray). Top: large β , bottom: small β (cf. C). Vertical lines mark sample points in B. (E) Average correct recall for different SNRs and β . Correct recall means the network has the largest overlap with the cued pattern after 1s (cf. D). Error bars indicate standard deviations over 7 randomly initialized networks, each storing 10 binary patterns. Each pattern was probed with 10 cues. (F) Average recall accuracy, quantified as the normalized distance between the largest and the average neural overlap with stored patterns after 1s. Error bars are standard deviations over recall cues ($n = 700$). (G) Anti-Hebbian plasticity recorded in CP-AMPA-deprived parvalbumin-positive interneurons and controls ($p = 0.03$, mixed ANOVA). Error bars are $\pm$ SEM. (H) Spatial memory task. To get a reward in the choice phase, animals had to recall and choose the T-maze arm that was blocked in the preceding sampling phase. (I) Animal performance depended on the delay between sampling and choice phase ($p = 0.001$, mixed ANOVA). Dashed line (gray) indicates performance for random policy. Error bars are $\pm$ SEM. (J) Recall performance of a model network reflecting experimental results in I. Data replotted from E. Error bars are $\pm$ SEM. (K) In the Morris water maze mice had to find a platform that was submerged in opaque water. After some time, the platform location was reversed to a different quadrant. (L) Time spent in target quadrant after learning the initial platform location does not differ between CP-AMPA-deprived (GluA2) and control animals ($p = 0.43$, one-way ANOVA). (M) Time spent in previous target quadrant ($p = 0.025$, mixed ANOVA). Error bars are $\pm$ SEM. (N) Network recall performance as a function of overlap between 2 out of 10 stored memory patterns. Average over 7 networks, 2 patterns per network, and 10 cues per pattern. Error bars are standard deviations over networks. Inset: two example patterns with 20% overlap.

to controls (Fig. 1I), matching our model's predictions (Fig. 1J). Similarly, in the water maze, CP-AMPA-deprived mice performed indistinguishably from controls when learning the initial platform location (Fig. 1L). However, when the platform was repositioned, animals spent more time in the *previous* target quadrant compared to controls (Fig. 1M), indicating difficulty in separating similar memory patterns. This was reflected in our networks with decreased anti-Hebbian synaptic strength, where re-

call performance dropped with increasing overlap between memory patterns (Fig. 1N).

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